

WEST FREUGH, United Kingdom

WMO: 031320

Lat: 54.8592N

Lon: 4.9353W

Elev: 12

StdP: 101.18

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.6	-2.0	-5.3	2.4	-0.8	-4.1	2.7	0.0	17.8	7.1	15.5	7.2	1.2	320	0.703

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.2	21.5	17.0	19.7	16.0	18.4	15.3	17.7	20.3	16.8	18.8	16.1	17.6	4.5	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.6	11.9	18.8	15.9	11.3	17.6	15.3	10.9	16.9	50.0	20.5	47.2	19.0	45.0	17.6	21.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.9	12.3	11.1	DB	-5.8	24.7	1.6	2.0	-7.0	26.2	-7.9	27.4	-8.8	28.5	-10.0	30.0	(4)
(5)			WB	-6.0	19.4	2.0	1.2	-7.4	20.2	-8.6	20.9	-9.7	21.6	-11.2	22.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	5.0	5.1	5.9	7.8	10.3	12.8	14.4	14.3	12.7	10.2	7.5	5.4	(6)	
	DBStd	4.31	2.82	2.63	2.51	2.39	2.58	2.01	1.96	1.86	2.14	2.64	2.89	3.34	(7)	
	HDD10.0	786	156	139	127	73	27	2	0	0	3	31	83	145	(8)	
	HDD18.3	3296	414	372	384	317	250	166	122	124	168	251	326	402	(9)	
	CDD10.0	532	0	1	1	7	36	86	138	134	85	38	7	1	(10)	
	CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0	(11)	
	CDH23.3	17	0	0	0	0	2	3	11	2	0	0	0	0	(12)	
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)	
(14) Wind	WSAvg	5.2	6.1	6.0	5.5	5.1	4.8	4.5	4.2	4.4	4.7	5.4	5.6	5.8	(14)	
(15) Precipitation	PrecAvg	977	101	63	71	61	55	60	67	90	100	106	106	97	(15)	
	PrecMax	1215	166	153	163	141	131	109	152	164	166	191	184	174	(16)	
	PrecMin	761	14	1	27	11	15	23	19	22	16	22	50	16	(17)	
	PrecStd	113	39	32	33	27	30	21	34	33	42	41	35	44	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	11.5	13.2	17.7	22.4	23.1	24.9	22.6	19.6	16.5	14.1	12.6	(19)	
		MCWB	10.3	10.2	10.1	12.2	15.6	16.7	18.9	18.2	16.4	14.4	13.1	11.6	(20)	
	2%	DB	10.7	10.6	11.7	14.5	18.9	20.0	21.8	20.1	18.0	15.5	13.2	11.7	(21)	
		MCWB	9.7	9.4	9.5	10.5	14.0	15.7	17.5	16.7	15.3	13.9	12.1	10.9	(22)	
	5%	DB	10.0	9.7	10.6	12.8	16.6	18.2	19.8	18.7	17.1	14.8	12.5	11.0	(23)	
		MCWB	9.1	8.6	8.8	9.8	12.8	14.8	16.4	15.8	14.8	13.4	11.4	10.1	(24)	
	10%	DB	9.2	9.0	9.8	11.6	14.8	16.9	18.3	17.7	16.3	14.2	11.8	10.1	(25)	
		MCWB	8.2	7.9	8.2	9.2	11.7	13.9	15.5	15.1	14.3	12.8	10.7	9.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	10.8	11.1	12.6	16.4	17.8	19.4	18.7	17.4	15.1	13.5	12.0	(27)	
		MCDB	11.2	11.2	12.3	16.3	20.6	21.4	23.6	21.5	18.8	15.8	13.9	12.3	(28)	
	2%	WB	10.0	9.6	10.0	11.3	14.7	16.4	18.0	17.5	16.2	14.5	12.4	11.1	(29)	
		MCDB	10.5	10.3	11.1	13.6	18.2	19.3	21.0	19.4	17.2	15.1	12.9	11.6	(30)	
	5%	WB	9.2	8.9	9.3	10.4	13.3	15.3	16.9	16.6	15.5	13.8	11.6	10.3	(31)	
		MCDB	9.8	9.6	10.2	12.1	15.8	17.7	19.3	18.0	16.5	14.6	12.3	10.9	(32)	
	10%	WB	8.4	8.1	8.6	9.7	12.2	14.4	16.0	15.9	14.7	13.1	10.9	9.3	(33)	
		MCDB	9.1	8.9	9.6	11.1	14.1	16.3	17.6	17.0	15.8	14.0	11.7	10.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.7	5.2	6.3	7.1	7.8	7.3	7.2	7.0	6.8	6.1	5.3	5.2	(35)	
		MCDBR	4.6	4.8	6.6	8.6	10.6	9.6	9.9	8.3	7.6	5.9	4.9	4.9	(36)	
	5% WB	MCWBR	4.6	4.3	5.0	5.8	6.9	6.2	6.3	5.5	5.4	4.9	4.8	5.1	(37)	
		MCWBR	4.6	4.4	5.9	7.4	9.2	8.9	8.6	6.7	6.3	5.0	4.7	5.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.334	0.345	0.369	0.386	0.390	0.384	0.384	0.370	0.363	0.348	0.327	0.320	(40)	
		taud	2.256	2.341	2.319	2.309	2.350	2.399	2.400	2.453	2.469	2.440	2.331	2.228	(41)	
	Ebn at Noon		581	730	798	837	855	864	855	847	803	722	593	504	(42)	
		Edh at Noon	60	77	98	113	115	110	108	97	84	69	55	50	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.54	1.22	2.40	3.87	4.96	4.96	4.75	3.90	2.71	1.47	0.70	0.40	(44)	
	RadStd		0.06	0.13	0.18	0.29	0.43	0.49	0.42	0.29	0.18	0.17	0.07	0.04	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page